JONATHAN ESPERANZA

jonathanesperanza67@gmail.com | linkedin/jonathanesperanza | github/jesperanzasolo

Applied AI engineer shipping production LLM systems: agent memory for 80M users, fine-tuned small models that replace frontier APIs, and tool-calling RAG agents.

EXPERIENCE

Yahoo Remote
Software Engineer, Personalization November 2025 - Present

- Built a self-serve **fine-tuning and RL platform for small language models** (vLLM, LoRA, GRPO), targeting tasks that don't need a frontier model.
- First production model replaced Gemini 2.5 Flash Lite for memory extraction at **2.1x throughput, 2.5x lower latency, and 2.1x lower cost per token**, while beating it on faithfulness (0.92 vs 0.79) and completeness (0.83 vs 0.68).
- Architected and own **Yahoo's Memory Platform for 80M users**, unifying Mail, Search, Finance, Shopping, and Scout signals into versioned, agent-ready profiles (confidence, decay, embeddings, entity mentions).
- Shipped production LLM memory extraction at **2000 events/sec**, writing 500 memories/sec, served to Yahoo agents through hybrid keyword and semantic retrieval.

Credit Karma Charlotte, NC
Software Engineer IV, Recommendations Platform (promoted from SWE I) August 2022 - November 2025

- Built a **RAG agent** that calls BigQuery tools over recommender logs to debug requests and answer on-call and support questions; cut debugging time an estimated 70%. **Won the company hackathon**; adopted as an internal tool.
- Built self-serve deployment of objective functions for deep-learning rankers, letting data scientists A/B test new ranking strategies in **hours instead of weeks.**
- Migrated real-time model serving to **NVIDIA Triton**: p99 latency 240ms to 180ms, and unblocked PyTorch and ONNX models in production.
- Re-architected ML training data from JSON to Parquet: storage cost down **25% ($850k/year)** and feature retrieval 48h to 6h; the faster retraining cadence won an A/B test at **+3% CTR**.
- Cut Cloud Dataflow cost **70% ($1.8M/year)** by replacing a large shuffle join in batch inference pipelines with a sort-merge bucket join.
- Owned batch ML pipelines generating **1.5B personalized notifications/month**, driving 24% of sitewide traffic.

EDUCATION

University of North Carolina at Charlotte Charlotte, NC
B.S Computer Science, Software Engineering August 2022

SKILLS

Programming Languages	Java, Scala, Python, Go, SQL
Cloud	Google Cloud Platform; Experience with AWS
Machine Learning	NVIDIA Triton, Pytorch, TensorFlow, ONNX, fine-tuning, reinforcement learning
LLM Systems	fine-tuning (LoRA), reinforcement learning (GRPO), vLLM, RAG, tool-calling agents, embeddings, hybrid search, LLM-as-judge evals, Gemini, Langfuse
Data	Apache Beam, Tensorflow, Apache Spark, Airflow, Parquet
Infrastructure	Kubernetes, Docker, Terraform
Certifications	Google Cloud Professional Machine Learning Engineer
Open Source	Apache Beam, Trino, Bento